

Spark Architecture, Driver & Executor

1. How does the Driver track Executor status?

2. Scenario: Driver has limited memory

Your Driver has **4 GB of memory** and you execute:

```
df.collect()
```

on a **500 GB dataset**.

- What happens?
- Why does it happen?
- How would you fix it?

3. How does Catalyst convert SQL into an execution plan?

4. Which Spark action is safest for large datasets?

5. Can Spark execute transformations in parallel?

Jobs, DAGs, Stages & Tasks

6. Scenario: Calling `show()` twice

```
df.show()
```

```
df.show()
```

- Will Spark execute the query twice?
- How can you avoid recomputation?

7. What is the role of the DAG Scheduler during job execution?

8. Can one Job belong to multiple Spark applications?

9. Can two Jobs share the same stages?

10. How do Jobs, Stages, and Tasks work together?

11. What is the difference between FIFO and FAIR scheduling?

12. You call `count()` twice on the same DataFrame. How many Jobs are created?

13. How does the DAG Scheduler recover from a failed stage?

14. What information is stored in the DAG?

15. How does the DAG Scheduler decide stage boundaries?

Caching & Persistence

16. Scenario: Cache and multiple actions

```
df.cache()  
df.count()  
df.show()
```

- Does the DAG change?
- Which action populates the cache?
- Will the second action/job reuse cached data?

17. How does DAG help improve performance in Spark jobs?

18. Can a DAG exist without RDD?

19. What is speculative execution?

20. How many times will Spark retry a failed task?

21. How does caching affect DAG execution?

22. When should you avoid caching?

23. Is cached data recomputed every time?

24. What is the BlockManager's role in caching?

Cluster Manager & Resource Management

- 25. What is the difference between a Cluster Manager and a Resource Manager?**
- 26. What is BlockManager?**
- 27. What is MemoryManager?**
- 28. Can an Executor communicate directly with another Executor?**
- 29. Does the Cluster Manager schedule Spark tasks?**
- 30. Compare Spark Standalone, YARN, Kubernetes, and Mesos as Cluster Managers.**
- 31. Can two Spark applications share the same Driver?**

Executor Memory

32. Scenario: Executor memory is full

- What happens to cached data?
- What happens to new tasks?

33. What is memory fragmentation in Spark?

34. Is Executor memory static or dynamic?

35. How does memory management work inside an Executor JVM?

36. How do you decide the number of Executors for a Spark job?

Partitions & Parallelism

37. You have 100 partitions and 10 cores. How many tasks run simultaneously?

38. If there are 1,000 cores but only 10 partitions, how many tasks run simultaneously?

39. How do you reduce task execution time?

40. One Executor has more tasks than other Executors. Why?

41. What happens if all Executors fail?

Driver–Executor Communication

- 42. Explain the network communication between Executors and the Driver.
- 43. How do Executors send results to the Driver?
- 44. How does heartbeat communication work between the Driver and Executors?

Failure Handling & Task Execution

- 45. What is speculative execution?
- 46. How many times will Spark retry a failed task?
- 47. How does the DAG Scheduler recover from a failed stage?
- 48. What happens if all Executors fail?

File Formats & Storage

- 49. How does ORC differ from Parquet internally?**
- 50. How does compression improve Spark performance?**
- 51. Why does Parquet reduce storage cost?**
- 52. Why does partitioning matter with file formats?**
- 53. Is compression always good for performance?**
- 54. How does Spark interact with Hadoop file formats?**
- 55. Why is partition pruning important in file formats?**